

# Md Mizanur Rahman, Ph.D.

## Forest & Ecosystem Ecologist | Global-Change Ecology | Carbon Stability & Resilience

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### RESEARCH PROFILE

Forest and ecosystem ecologist investigating how biodiversity, functional traits, ecosystem structure, climate, hydrology and disturbance regulate ecosystem functioning, carbon stability and resilience across space and time. I integrate long-term field ecology with Earth observation and quantitative modelling across tropical and subtropical forests, wetlands, agroforestry systems and other natural and managed landscapes. My current flagship research quantifies the temporal stability of pantropical aboveground forest carbon (2000-2025) under compound heat, drought and wet extremes and tests whether carbon-stock stability and productivity stability decouple across climatic and ecological gradients. My first-author research in Nature Communications, Global Change Biology and Communications Earth & Environment has advanced understanding of biodiversity-carbon relationships, functional controls on ecosystem processes and forest resilience. Earth observation, UAV/LiDAR, machine learning/GeoAI and structural equation modelling serve as tools to test ecological mechanisms from plots to landscapes and long time series.

### RESEARCH EXPERTISE

**Science:** global-change ecology; forest resilience; carbon stability and vulnerability; biodiversity-ecosystem functioning; functional traits and composition; climate x biodiversity interactions; productivity; hydrology; disturbance, recovery and restoration; Nature-based Solutions.

**Methods:** field ecology and carbon inventory; UAV/TLS LiDAR; Landsat, Sentinel-1/2, MODIS, GEDI, WorldView-2, TanDEM-X; Google Earth Engine; R and Python; machine learning/GeoAI; spatial and time-series analysis; structural equation modelling.

### SELECTED RESEARCH CONTRIBUTIONS

- Biodiversity-carbon co-benefits - demonstrated that mangrove protection can deliver joint biodiversity conservation and blue-carbon benefits (Nature Communications, 2021).
- Mechanisms of ecosystem functioning - showed that functional composition strongly regulates multiple ecosystem properties and processes, with species richness buffering environmental stress (Global Change Biology, 2024).
- Forest resilience - integrated 25 years of Earth observation with field biodiversity, traits, structure, climate and disturbance to identify biological and structural controls on post-disturbance recovery (Communications Earth & Environment, 2026).
- Ecological remote sensing - mapped canopy structure, functional traits and carbon using WorldView-2/TanDEM-X, Landsat/machine learning, Sentinel, GEDI, UAV and terrestrial LiDAR; contributed to the 30 m global mangrove soil-carbon map.
- Current frontier - lead a pantropical analysis of aboveground forest-carbon temporal stability under compound climate extremes, linking carbon trajectories with productivity, biodiversity, functional traits, canopy structure, soils, hydrology and disturbance.

### EDUCATION

|                                                                         |      |
|-------------------------------------------------------------------------|------|
| Ph.D., Forest and Biomaterials Science - Kyoto University, Kyoto, Japan | 2019 |
| M.Sc., Forestry - Khulna University, Khulna, Bangladesh                 | 2012 |
| B.Sc. (Hons.), Forestry - Khulna University, Khulna, Bangladesh         | 2009 |

### RESEARCH AND PROFESSIONAL APPOINTMENTS

#### Research Fellow

Jun 2026-present

Remote Sensing Laboratory, Department of Land Surveying and Geospatial Science, The Hong Kong Polytechnic University, Hong Kong SAR, China

- Lead pantropical research on temporal stability of aboveground forest carbon (2000-2025) under compound climate extremes, integrating productivity, biodiversity, functional traits, canopy structure, soils, hydrology and disturbance.
- Investigate hydrological controls on ecosystem and soil carbon in Hong Kong coastal ecosystems and contribute to mangrove blue-carbon research.
- Support tree-growth and urban green-space carbon monitoring for Hong Kong and PolyU carbon-management initiatives.
- Integrate field observations with UAV/TLS LiDAR, Sentinel, Landsat, MODIS and GEDI to quantify ecosystem structure, carbon dynamics and resilience.

#### Research Fellow

Mar 2024-Jun 2026

Jockey Club STEM Laboratory of Earth Observations, Department of Land Surveying and Geo-Informatics (now LSGS), The Hong Kong Polytechnic University, Hong Kong SAR, China

- Developed a 25-year remote-sensing and field framework for forest resilience, culminating in a first-author Communications Earth & Environment paper (2026).
- Led wetland-group field research using UAV LiDAR and terrestrial LiDAR to assess forest structure, biodiversity and carbon in Hong Kong mangroves.
- Chaired a weekly interdisciplinary seminar series on geospatial technologies for socioecological and environmental challenges.

#### Assistant Researcher

May 2023-Feb 2024

Carbon Accounting and Carbon Resource Utilization, Jiangmen Laboratory of Carbon Science and Technology / HKUST(GZ), Guangdong, China

- Advanced ecosystem-carbon assessment for forest-based climate mitigation and carbon accounting.
- Integrated drone and airborne remote sensing with machine learning/deep learning for spatial and temporal monitoring of forest carbon and functional diversity.

**RGC Postdoctoral Fellow****Jan 2022-May 2023***School of Biological Sciences, The University of Hong Kong, Hong Kong SAR, China*

- Integrated Sentinel-1/2, GEDI, TanDEM-X and Landsat data with machine learning to monitor forest biophysical properties.
- Investigated drivers of multiple ecosystem properties and processes and biodiversity-environment interactions.

**Postdoctoral Fellow****Feb 2019-Dec 2021***College of Geography and Environmental Science, Henan University, Kaifeng, China*

- Led biodiversity-carbon and functional-trait research using field observations, structural equation modelling and Landsat time series.
- Developed research that produced first-author papers in Nature Communications, Global Change Biology and related journals.

**Assistant Research Staff****Nov 2018-Jan 2019***Division of Forest and Biomaterials Science, Kyoto University, Kyoto, Japan*

- Applied photogrammetry and WorldView-2 stereo imagery to forest canopy mapping.

**NASA Visiting Fellow****Jul-Sep 2016***NASA Goddard Space Flight Center, Greenbelt, MD, USA*

- Mapped forest canopy height and 3-D vegetation structure using WorldView-2 and TanDEM-X.

**Junior Professional****Feb 2014-Aug 2015***Remote Sensing Division, Center for Environmental and Geographic Information Services, Bangladesh*

- Led permanent-plot forest-health inventories in the Sundarbans; conducted land-use classification and environmental-impact assessments; contributed to proposals and technical reports.

**Research Assistant****Jan 2013-Jan 2014***Department of Geography, Indiana University, Bloomington, IN, USA / Bangladesh*

- Collected plant functional traits; led carbon inventory in the former Chakaria Sundarbans; assessed restoration options and stakeholder perspectives.

**Field Assistant****Aug 2011-Dec 2012***Forestry and Wood Technology Discipline, Khulna University, Bangladesh*

- Supported agroforestry promotion, socioeconomic surveys, plantation monitoring and student research.

**Research Assistant****Jun 2010-Jul 2011***Forestry and Wood Technology Discipline, Khulna University, Bangladesh*

- Led field inventory teams assessing biodiversity and stand structure in the Sundarbans.

**Carbon Inventory Analyst****Dec 2009-May 2010***USAID / Bangladesh Forest Department, Bangladesh*

- Collected and analysed forest-carbon, land-use and socioeconomic data across mangrove and tropical forest systems.

**SELECTED HONORS, FELLOWSHIPS AND RESEARCH FUNDING**

|                                                                                                                       |           |
|-----------------------------------------------------------------------------------------------------------------------|-----------|
| Hong Kong RGC Postdoctoral Fellowship Scheme - Distinguished Postdoctoral Fellowship                                  | 2022-2023 |
| Henan Provincial Government research grant - Principal Investigator (approx. US\$12,246)                              | 2021-2022 |
| College of Geography and Environmental Science Performance Award, Henan University                                    | 2021      |
| NASA/SilvaCarbon Visiting Fellowship - ranked 1st among approximately 400 applicants in Bangladesh                    | 2016      |
| Kyoto University travel award - Mangrove and Macroinvertebrates Meeting, Florida                                      | 2016      |
| MEXT (Monbukagakusho) Scholarship for Ph.D., Government of Japan                                                      | 2015-2019 |
| Canopy height and carbon stocks assessment in Bangladesh - Co-PI, German Aerospace Center / NASA-hosted collaboration | 2017-2019 |
| Environmental monitoring projects - Team Leader, Sundarbans forest-health monitoring, CEGIS                           | 2014-2015 |
| Tree carbon and agroforestry research - Co-PI, University Grants Commission, Bangladesh                               | 2013-2014 |
| Roadside plantation biomass-carbon project - Khulna University Research Cell                                          | 2012-2013 |

**PEER-REVIEWED PUBLICATIONS***Google Scholar (September 2026): 1,550 citations | h-index 14 | i10-index 19.*

1. Cao, M., Mohammad, P., **Rahman, M. M.**, & Weng, Q. (2026). Differential impacts of soil hydrothermal properties on root and leaf phenology in cropland. *International Journal of Applied Earth Observation and Geoinformation*, 149, 105282. <https://doi.org/10.1016/j.jag.2026.105282>
2. **Rahman, M. M.**, Zimmer, M., Rahman, S. M., Liang, J., Khan, N. I., Donato, D., & Weng, Q. (2026). Functional composition and structural diversity enhance mangrove forest resilience in the Sundarbans. *Communications Earth & Environment*. <https://doi.org/10.1038/s43247-026-03305-5>
3. Ahmed, S., Sarker, S. K., Kamruzzaman, M., Perri, S., Hilmers, T., Uhl, E., Hossain, M. R., Tasnim, N., Naabeh, C. S. S., Anika, T. T., **Rahman, M. M.**, & Pretzsch, H. (2025). Soil nutrients and leaf area index interact with species and structural diversity to buffer mangrove productivity against salinity. *Soil Ecology Letters*, 7, 250299.
4. Wang, J., Li, Y., **Rahman, M. M.**, Wang, W., Wu, J., & Chu, C. (2024). Characterizing phenological diversity and associated drivers in a subtropical evergreen forest using CubeSats observations. *New Phytologist*. <https://doi.org/10.1111/nph.19850>
5. **Rahman, M. M.**, Zimmer, M., Donato, D., Ahmed, I., Xu, M., & Wu, J. (2024). Functional composition outweighs taxonomic and functional diversity in maintaining ecosystem properties and processes of mangrove forests. *Global Change Biology*, 30, e17152.

6. **Rahman, M. M.**, Zimmer, M., Ahmed, I., Donato, D., Kanzaki, M., & Xu, M. (2021). Co-benefits of protecting mangroves for biodiversity conservation and carbon storage. *Nature Communications*, 12, 3875.
7. **Rahman, M. M.**, Kundu, G. K., Ahmed, H., Kabir, M. E., & Xu, M. (2021). Opposing ecological strategies together promote biomass carbon storage in homegarden agroforestry of southern Bangladesh. *Forests*, 12, 1669.
8. **Rahman, M. M.**, Kundu, G. K., Ahmed, H., Kabir, M. E., & Xu, M. (2021). Assessing tree coverage and the direct and mediation effect of tree diversity on carbon storage through stand structure in homegardens of southwestern Bangladesh. *Forests*, 12, 1661.
9. Zhang, X., Zeraatpisheh, M., **Rahman, M. M.**, Wang, S., & Xu, M. (2021). Texture is important in improving the accuracy of mapping photovoltaic power plants: A case study of Ningxia Autonomous Region, China. *Remote Sensing*, 13, 3909.
10. **Rahman, M. M.**, Zhang, X., Ahmed, I., Iqbal, Z., Zeraatpisheh, M., Kanzaki, M., & Xu, M. (2020). Remote sensing-based mapping of senescent leaf C:N ratio in the Sundarbans Reserved Forest using machine learning techniques. *Remote Sensing*, 12, 1375.
11. **Rahman, M. M.**, Lagomasino, D., Lee, S. K., Fatoyinbo, T. E., Ahmed, I., & Kanzaki, M. (2019). Improved assessment of mangrove forests in Sundarbans East Wildlife Sanctuary using WorldView-2 and TanDEM-X high-resolution imagery. *Remote Sensing in Ecology and Conservation*, 5, 136-149.
12. Sanderman, J., et al. (including **Rahman, M. M.**) (2018). A global map of mangrove forest soil carbon at 30 m spatial resolution. *Environmental Research Letters*, 13, 055002.
13. Kamruzzaman, M., Ahmed, S., Paul, S., **Rahman, M. M.**, & Osawa, A. (2018). Stand structure and carbon storage in the oligohaline zone of the Sundarbans mangrove forest, Bangladesh. *Forest Science and Technology*, 14, 23-28.
14. **Rahman, M. M.**, **Rahman, M. M.**, & Rahman, M. (2018). Environmental quality evaluation in Dhaka City Corporation using satellite imagery. *Urban Design and Planning*. <https://doi.org/10.1680/jurdp.17.00032>
15. **Rahman, M. M.**, Kabir, M. E., & Ahmed, I. (2017). Protected areas for climate change mitigation and livelihood option: A case study of the Bangladesh Sundarbans mangrove forest. In R. DasGupta & R. Shaw (Eds.), *Participatory Mangrove Management in a Changing Climate* (pp. 119-136). Springer.
16. **Rahman, M. M.**, Khan, M. N. I., Hoque, A. K. F., & Ahmed, I. (2015). Carbon stock in the Sundarbans mangrove forest: Spatial variations in vegetation types and salinity zones. *Wetlands Ecology and Management*, 23, 269-283.
17. **Rahman, M. M.**, Kabir, M. E., Akon, A. S. M. J. U., & Ando, K. (2015). High carbon stocks in roadside plantations under participatory management in Bangladesh. *Global Ecology and Conservation*, 3, 411-423.
18. **Rahman, M. M.**, Das, A. K., Asaduzzaman, M., Biswas, S. K., & Hannan, M. O. (2014). Physical and mechanical properties of Raj Koro (Albizia richardiana) plywood. *African Journal of Wood Science and Forestry*, 2, 98-103.
19. Shanu, S. A., Das, A. K., **Rahman, M. M.**, & Ashaduzzaman, M. (2015). Effect of CCB preservative treatment on physical and mechanical properties of Raj Koro (Albizia richardiana) wood. *Bangladesh Journal of Scientific and Industrial Research*, 50, 77-86.
20. Alam, J., Das, A. K., **Rahman, M. M.**, & Islam, A. (2015). Effect of waterlogged condition on wood properties of *Acacia nilotica*. *Bangladesh Journal of Scientific and Industrial Research*, 50, 71-76.
21. Ghosh, R. K., **Rahman, M. M.**, Das, A. K., Rana, R., & Shams, M. I. (2015). Introducing Areca catechu as a raw material of cement-bonded board through determining the properties of Areca catechu cement-bonded board. *Journal of the Indian Academy of Wood Science*, 12, 99-103.
22. Rahman, K., Asaduzzaman, M., **Rahman, M. M.**, Das, A. K., & Biswas, S. K. (2014). Physical and mechanical properties of Ghora neem (*Melia azedarach*) plywood. *Bangladesh Journal of Scientific and Industrial Research*, 49, 47-52.
23. Zaman, M. S., Das, A. K., Shams, M. I., Rana, R., **Rahman, M. M.**, & Biswas, S. K. (2013). Physical and mechanical properties of agricultural plastic composite made from jute (*Corchorus olitorius*) and dhaincha (*Sesbania cannabina*). *International Journal of Wood Science Technology and Forestry*, 1, 1-8.

## CURRENT RESEARCH AND MANUSCRIPTS IN DEVELOPMENT

- Pantropical forest carbon stability under compound climate extremes - lead analysis integrating annual carbon trajectories (2000-2025), productivity, biodiversity, functional traits, canopy structure, soils, hydrology and disturbance.
- Remnant-first mangrove restoration in aquaculture landscapes - synthesis on ecological memory, propagule sources, additionality and restoration safeguards.
- Agroforestry as a Nature-based climate solution in the Indo-Gangetic Plain - collaborative analysis of land-use dynamics, carbon and ecosystem function.

## TEACHING, TRAINING AND MENTORING

- Guest Lecturer, Forest Biodiversity and Ecosystem Functioning, School of Biological Sciences, The University of Hong Kong (2022).
- Teaching Assistant, mangrove carbon mapping in Myanmar using Sentinel-1/2, Kyoto University (2018).
- Teaching Assistant, forest-cover change in Indonesian mangroves using Landsat and Sentinel-1/2, Kyoto University (2017).
- Trainer, Intermediate Remote Sensing, GIS and GPS for Bangladesh Forest Department, US Forest Service International Program / CEGIS (2014).
- Trainer, SAARC Workshop on Measurement and Estimation of Carbon Stocks in Mangrove Forests of South Asia, participants from eight South Asian countries (2011).
- Supervised, co-supervised or mentored 14+ undergraduate, postgraduate and early-career researchers in forest carbon, biodiversity, agroforestry, UAV/LiDAR, remote sensing, ecological modelling and scientific writing.
- At PolyU, mentor students and early-career researchers in field ecology, terrestrial/UAV LiDAR, ecological theory, structural equation modelling, data visualization and manuscript development.

## SELECTED RESEARCH PRESENTATIONS

|                                                                                                                                               |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------|
| Predictability of Landsat TM 5 for leaf C:N ratio in the Sundarbans: machine-learning regression, ECOSUMMIT 2020/2022 (online)                | 2022 |
| Carbon stocks, vegetation type, tree diversity and canopy height - USAID Office, Dhaka                                                        | 2017 |
| Carbon stocks, vegetation type, tree diversity and canopy height - Bangladesh Forest Department                                               | 2017 |
| Carbon stocks in relation to vegetation type, tree diversity and tree-height class - NASA Goddard Space Flight Center                         | 2016 |
| Carbon stocks in relation to vegetation type, tree diversity and tree-height class - US Forest Service International Programs, Washington, DC | 2016 |
| Biomass carbon stocks in protected areas in Chittagong Hills - 127th Annual Meeting of the Japanese Forest Society                            | 2016 |

## PROFESSIONAL CERTIFICATIONS AND TECHNICAL SKILLS

**Registered Remote Pilot - Advanced Operations, Hong Kong:** Authorized to conduct advanced UAV operations in Hong Kong; current certification valid through 2028.

**Field ecology & carbon inventory:** Forest and soil carbon protocols; permanent plots; biodiversity, stand structure and functional-trait measurements; GPS/DGPS and standard inventory equipment.

**UAV and LiDAR:** UAV photogrammetry, UAV LiDAR, hyperspectral/multispectral surveys, terrestrial laser scanning and point-cloud processing.

**Earth observation:** Landsat, Sentinel-1/2, MODIS, GEDI, WorldView-2, TanDEM-X; Google Earth Engine, SNAP, ENVI, ArcGIS and QGIS.

**Quantitative ecology & modelling:** R, Python, structural equation modelling, time-series analysis, spatial statistics, random forest, XGBoost, GBM, SVM and interpretable ML/GeoAI.

## ACADEMIC SERVICE

- Topical Advisory Panel, Remote Sensing (2021-present).
- Topic Editor, Remote Sensing (2021).
- Peer reviewer: Journal of Applied Ecology; Regional Environmental Change; Remote Sensing; Remote Sensing in Ecology and Conservation; Carbon Management; Forests; Ecosystem Health and Sustainability; Wetlands Ecology and Management; Journal of Sustainable Forestry; Small-scale Forestry.
- Chair, weekly interdisciplinary guest-speaker seminar series at PolyU on geospatial technologies for environmental and socioecological challenges (2024-2026).

## PROFESSIONAL MEMBERSHIPS AND LANGUAGES

**Memberships:** The Japanese Forest Society; International Biogeography Society (former member).

**Languages:** Bengali (native); English (fluent); Hindi (basic communication); Japanese (basic); Chinese (very basic).

## REFERENCES

**Prof. Mamoru Kanzaki** - Professor Emeritus, School of Agriculture, Kyoto University, Japan | kanzaki.mamoru.j90@kyoto-u.jp

**Dr. Daniel Donato** - Natural Resource Scientist, Washington Department of Natural Resources; Affiliate/Assistant Professor, University of Washington, USA | Daniel.Donato@dnr.wa.gov

**Prof. Martin Zimmer** - Leibniz Centre for Tropical Marine Research (ZMT) and University of Bremen, Germany | martin.zimmer@leibniz-zmt.de

**Prof. Ming Xu** - Deputy Managing Director, Jiangmen Laboratory of Carbon Science and Technology; Director, BNU-HKUST Laboratory of Green Innovation | mingxu@henu.edu.cn

**Prof. Qihao Weng** - Chair Professor of Geomatics and Artificial Intelligence; Director, Research Centre for Artificial Intelligence in Geomatics; Associate Director, Research Institute for Land and Space, The Hong Kong Polytechnic University, Hong Kong SAR | qihao.weng@polyu.edu.hk

**Prof. Md. Nabiul Islam Khan** - Forestry and Wood Technology Discipline, Khulna University, Bangladesh | mnikhan@fwt.ku.ac.bd